7 Two coins are projected from a horizontal table at the same initial speed u.

Coin X is projected horizontally.

Coin Y is projected upwards at an angle of 30° to the horizontal.

Both coins hit the horizontal ground without bouncing.

Assume the air resistance on each coin is negligible.

Which statement about the motion of the coins is correct?

A Both coins hit the ground with the same speed.

B Both coins travel the same vertical distance.

C Coin Y hits the ground before coin X.

D Coin Y has a smaller vertical acceleration.